Genetic Disorders

Worksheet · PU-II Biology, Chapter 4

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Worksheet – Genetic Disorders

15 questions · answer key on the last page

Q1. Genetic disorders are broadly classified into:

(a) Autosomal and sex-linked disorders only
(b) Mendelian disorders and Chromosomal disorders
(c) Dominant and recessive disorders only
(d) Physical and chemical disorders

Q2. Mendelian disorders are caused by:

(a) Changes in chromosome number
(b) A mutation in a single gene, following Mendelian inheritance principles
(c) Environmental factors only
(d) Multiple genes acting cumulatively

Q3. Which of the following is an example of an autosomal dominant disorder?

(a) Sickle cell anaemia
(b) Huntington disease
(c) Phenylketonuria
(d) Cystic fibrosis

Q4. Which of the following is an example of an autosomal recessive disorder?

(a) Huntington disease
(b) Sickle cell anaemia
(c) Duchenne muscular dystrophy
(d) Haemophilia

Q5. Haemophilia is inherited as a(n):

(a) Autosomal dominant trait
(b) Autosomal recessive trait
(c) X-linked recessive trait
(d) Y-linked trait

Q6. Duchenne muscular dystrophy is inherited as a(n):

(a) X-linked recessive trait
(b) X-linked dominant trait
(c) Autosomal recessive trait
(d) Y-linked trait

Q7. Hypertrichosis (hairs on the pinna of the ear) is an example of a:

(a) X-linked recessive trait
(b) X-linked dominant trait
(c) Y-linked trait
(d) Autosomal dominant trait

Q8. Down's syndrome is caused by:

(a) Deletion of part of chromosome 5
(b) An extra copy of chromosome 21 (trisomy 21)
(c) Absence of one X chromosome
(d) An extra X chromosome (XXY)

Q9. Klinefelter's syndrome is caused by which chromosomal constitution?

(a) 45, XO
(b) 47, XXY

(c) 46, XY
(d) 47, XYY

Q10. Turner's syndrome is caused by which chromosomal constitution?

(a) 47, XXY
(b) 45, XO
(c) 46, XX
(d) 47, XXX

Q11. Aneuploidy involving the loss of one chromosome from a homologous pair (2n-1) is called:

(a) Trisomy
(b) Monosomy
(c) Polyploidy
(d) Euploidy

Q12. Sickle cell anaemia results from a base substitution on chromosome 11 that causes which amino acid change in the beta globin chain?

(a) Valine replaced by Glutamic acid at position 6
(b) Glutamic acid replaced by Valine at position 6
(c) Glycine replaced by Alanine at position 6
(d) Leucine replaced by Proline at position 6

Q13. The mutant sickle-cell haemoglobin polymerises and distorts the RBC into a sickle shape under conditions of:

(a) High oxygen tension
(b) Low oxygen tension
(c) High glucose levels
(d) Low body temperature

Q14. Thalassemia is best described as a disease resulting from:

(a) An excess of normal haemoglobin
(b) A quantitatively deficient (reduced amount of) haemoglobin, due to a defective globin gene
(c) A completely absent circulatory system
(d) An extra chromosome 21

Q15. Alpha thalassemia is controlled by genes located on which chromosome, according to the source material?

(a) Chromosome 11
(b) Chromosome 12
(c) Chromosome 16
(d) Chromosome 7

Answer Key & Explanations

Q1. (b) Mendelian disorders and Chromosomal disorders — Genetic disorders are broadly classified into Mendelian disorders, caused by mutation in a single gene, and Chromosomal disorders, caused by changes in chromosome number or structure.

Q2. (b) A mutation in a single gene, following Mendelian inheritance principles — Mendelian disorders arise from mutation in a single gene and follow the classic principles of Mendelian inheritance.

Q3. (b) Huntington disease — Huntington disease (on chromosome 4) is an autosomal dominant disorder, while sickle cell anaemia, PKU, and cystic fibrosis are autosomal recessive.

Q4. (b) Sickle cell anaemia — Sickle cell anaemia is an autosomal recessive disorder, unlike Huntington disease (autosomal dominant) or Duchenne muscular dystrophy and haemophilia (both sex-linked).

Q5. (c) X-linked recessive trait — Haemophilia is an X-linked recessive disorder, caused by a defect in the gene for clotting Factor VIII.

Q6. (b) X-linked dominant trait — Duchenne muscular dystrophy is an X-linked dominant disorder, in contrast to haemophilia and colour blindness which are X-linked recessive.

Q7. (c) Y-linked trait — Hypertrichosis is a Y-linked trait, so it is transmitted only from father to son and never appears in females.

Q8. (b) An extra copy of chromosome 21 (trisomy 21) — Down's syndrome, first described by Langdon Down in 1866, is caused by an extra copy of chromosome 21, i.e. autosomal trisomy 21.

Q9. (b) 47, XXY — Klinefelter's syndrome results from an additional X chromosome, giving a chromosomal constitution of 47, XXY, an allosomal trisomy.

Q10. (b) 45, XO — Turner's syndrome results from the absence of one X chromosome, giving a chromosomal constitution of 45, XO, an allosomal monosomy.

Q11. (b) Monosomy — Monosomy (2n-1) is the loss of one chromosome from a homologous pair, while trisomy (2n+1) is the gain of an extra chromosome.

Q12. (b) Glutamic acid replaced by Valine at position 6 — In sickle cell anaemia, a GAG to GUG substitution causes Glutamic acid to be replaced by Valine at the sixth position of the beta globin chain.

Q13. (b) Low oxygen tension — Mutant sickle-cell haemoglobin polymerises under low oxygen tension, causing the red blood cell to become sickle-shaped.

Q14. (b) A quantitatively deficient (reduced amount of) haemoglobin, due to a defective globin gene — Thalassemia results from a defective globin gene causing the body to make a quantitatively deficient amount of haemoglobin, leading to excessive RBC destruction and anaemia.

Q15. (c) Chromosome 16 — Alpha thalassemia is caused by defects in two closely linked genes, HBA1 and HBA2, located on chromosome 16.